



Crop Insurance Data Analysis Dashboard

Comprehensive Data Cleaning & Analysis Report (2018-2022)



Dataset Overview

6,161

Original Records

5,694

Clean Records

28

Total Columns

7.6%

Data Reduction



Data Quality Assessment



Issues Identified

Missing Values: District codes missing in ~12% of rows

Invalid Entries: Zero/infinite values in financial metrics

Data Inconsistency: 11 rows with negative demographic percentages

Count Discrepancy: ~35,169 avg difference in farmer counts



Data Integrity Checks

Premium Share Consistency: farmer_share + goi_share + state_share ≈ gross_premium ($\sigma \approx 0.00$)

Preserved Missing Data: 70 rows with missing demographics kept for non-participation analysis



7-Step Data Cleaning Process

1

Remove Invalid Demographics

Dropped 10 rows with negative demographic percentages

2

Preserve Missing Demographics

Kept 70 rows with missing demographics for non-participation pattern analysis

3

Create Consistent Policy Count

Created total_policies = loanee + non_loanee for reliable metrics

4

Remove Logical Inconsistencies

Dropped 457 rows with logical errors (e.g., policies > 0 but sum_insured = 0)

5

Clip Demographic Percentages

Constrained demographic percentages to [0, 100] range

6

Fix Data Types

Converted district_code to Int64, standardized year and state_code types

7

Reset Index

Final cleanup step for consistency



Statistical Analysis Results

Key Metrics Summary

Metric	Mean	Max	Skewness
Area Insured	44.8 ha	3,777 ha	11.36
Sum Insured	₹17,305	₹5.35 lakh	3.8+
Gross Premium	₹2,441	₹80,103	5.25
% Transgender	-0.001	4.9	42.34



Key Findings

- All financial metrics show heavy right-skew distribution
- Strong correlation between sum_insured ↔ premium ↔ policies
- 2018 was the most common year in dataset
- High variance in total_policies suggests need for log-scaling
- Demographics show tight distributions with few outliers
- Extreme outlier behavior in transgender percentage data



Demographic Analysis

Gender Distribution

Male: 86.0% | Female: 13.9% | Transgender: 0.1%

Strong male dominance in crop insurance participation

Caste Distribution

Gen: 45.3% | OBC: 38.6% | ST: 10.2% | SC: 5.9%

General and OBC categories dominate participation

Farmer Type Distribution

Small: 60.7% | Marginal: 21.5% | Other: 17.8%

Small farmers are the primary beneficiaries

Financial Distribution

Gross Premium & Area Insured show heavy right-skew

Most policies are small-scale with few large outliers



Feature Classification

Numerical (Continuous)

20

Used for descriptive stats and histograms

Numerical (Discrete)

5

Count variables (e.g., total_policies)

Categorical (Nominal)

6

Used for grouping and bar plots



Final Data Quality Summary

Cleaning Results



Final Dataset: 5,694 clean records ready for analysis



Data Reduction: 7.6% of problematic records removed



Output File: cleaned_crop_insurance1.csv

Ready for Modeling



Log-scaling: Recommended for high-variance variables



Correlations: Strong relationships identified for feature engineering



Visualizations: All charts created with proper labeling